Lappeenrannan teknillinen yliopisto
School of Software Engineering

Sofware Development Skills Full Stack

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LEARNING DIARY, FULL-STACK MODULES

LEARNING DIARY

NodeJS

6.8.2026

I started with the Node.js tutorial video and coded along with it. I have already used node and java script, but it was nice to freshen my memory on it. In the process of fiddling with these tools I learned some interesting facts that I might have heard before but forgotten. For example Node is a javascript runtime that is built on chromes V8 engine which allows it to run outside the browser. There is no DOM instead I get built in modules such as fs, path and os. I learned about CommonJS modules and made a webserver using the http module and read req.url for manual routing.

MongoDB

10.8.2026

I started my work on the MongoDB video. I had again previously used MongoDB but somehow it seemed to be broken so I did a fresh install and proceeded along. I learned that MongoDB differs from relative databases in that it uses BSON which stores data in collections instead of rows in tables. I used the Shell and Compass in order to run commands like insertOne, insertMany, find, updateOne with $set, and deleteOne. I also queried filter operators such as $gt and $in.

ExpressJS

15.8.2026

I had used ExpressJS before but as previously I started with the video and got a nice chance to refresh my memory. I learned that express is an additional layer on top of nodes http

module. I made routes with app.getm read route parameters with query strings, sent JSON with res.json and served a static folder with express.static.
I also refreshed my memory about middleware: functions with req, res and next that run in order before the route handler. I made a loggermiddleware, used express.json and urlencoded to read request bodies, and split the routes into separate files with express.Router. I tested POST, PUT, and DELETE requests with Postman. I also learned to keep settings like the port in .env.

React

21.8.2026

As previously I started by opening up the video and my VsCode and got to work. I made a React project with Vite and learned things such as JSX, props and rendering lists with a map and a unique key. I learned the useState hook, event handlers, controlled form inputs and lifting state up inorder for sibling components to share data. I also learned useEffect and used it to fetch tasks.

MERN stack

20.9.2026 - 1.10.2026

I had trouble coming up with something interesting to do as my project, but after a long break I had a revelation to make a software that I could possibly use in the future in a personal network attached server kind of setup. I'm satisfied with the result in which I combined the knowledge from the previous modules into one project.
On the backend I made three Mongoose schemas: User, Folder, and File. I learned how to model relationships in MongoDB using ObjectId references such as owner and parentFolder. I built an auth system with JWT and bcrypt hashing passwords on register, signing a token on login, and writing an auth middleware that reads the authorization header and attaches the user to req. I also learned that <img> and <video> tags can’t send custom headers, so for file previews I had to let the same middleware accept the token as a query parameter too.

I used multer with diskStorage to hande multipart form uploads and generate unique filenames, and sharp to generate image thumbnails on the server. One of the most interesting parts was implementing video streming properly, I had to handle the Range header and respong with HTTP 206 partial content so the browser can actually seek through a video instead of only playing it from the start. I also added a per user storage quota that is checked against the total size of an upload before accepting it.

On the frontend I used React with Vite. In the app I have AuthContext using the Context API so login state is available everywhere, axios with an interceptor to attach the JWT to every request automatically, and React Router to protect routes so logged out users get redirected to login. I built the UI out of components such as: Sidebar, Toolbar, FileView and PreviewModal that all share state through one parent component. I also used native HTML5 drag and drop events, both for uploading files by dragging them into the window and for moving files between folders.